

Deepak Sathyan

Publication List

Publications

5. **New laboratory constraints on neutrinophilic mediators** July 2024
P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
arxiv.org/abs/2407.12738 (Phys.Lett.B 868 (2025) 139765)
4. **‘Unification’ of BSM searches and SM measurements: the case of lepton+MET and m_W** Apr 2024
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
arxiv.org/abs/2404.17574 (JHEP 02 (2025) 139)
3. **A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET** Oct 2023
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
arxiv.org/abs/2310.13687 (Phys.Lett.B 855 (2024) 138774)
2. **Energy-peak based method to measure top quark mass via B-hadron decay lengths** Dec 2022
K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan
arxiv.org/abs/2212.03929 (JHEP 06 (2023) 021)
1. **LHC Signals for KK Graviton from an Extended Warped Extra Dimension** Aug 2020
K. Agashe, M. Ekhterachian, D. Kim, D. Sathyan
arxiv.org/abs/2008.06480 (JHEP 11 (2020) 109)

Preprints

4. **Intensity-Frontier Signals of Warped Extra Dimensions** Aug 2026
D. Kim, D. Sathyan, A. Verma
arxiv.org/abs/2608.20482
3. **Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering** May 2026
B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan
arxiv.org/abs/2605.08010
2. **A Baryon and Lepton Number Violation Model Testable at the LHC** Aug 2025
A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan
arxiv.org/abs/2508.21064
1. **New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis** July 2025
P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
arxiv.org/abs/2507.01000

Articles in Preparation

1. LHC as a Beam Dump: Probing Light Dark Photons

B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, D. Sathyan

(In preparation)

2. New Probes of Dark Sectors at a Muon Collider

B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, D. Sathyan

(In preparation)

3. Exploring open systems approach to neutrino oscillations

A. Chandra Shekar, K. Kelly, D. Sathyan, L. Strigari, T. Zhou

(In preparation)

4. LHC Signals for KK Gravitons Producing 8-Particle Final States

K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, D. Sathyan

(In preparation)

Snowmass2021 Contributions

5. TF07 Snowmass Report: Theory of Collider Phenomena

Oct 2022

F. Maltoni et al.

arxiv.org/abs/2210.02591

4. Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021

Sept 2022

K. Agashe et al.

arxiv.org/abs/2209.11267

3. Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021

Sept 2022

T. Bose et al.

arxiv.org/abs/2209.13128

2. Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond

Apr 2022

K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan

arxiv.org/abs/2204.02928

1. Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models

Mar 2022

K. Agashe, J.H. Collins, P. Du, M. Ekhaterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan

arxiv.org/abs/2203.13305

Muon g-2 Articles

3. The straw tracking detector for the Fermilab Muon g-2 Experiment

Nov 2021

B.T. King et al.

arxiv.org/abs/2111.02076 (JINST 17 (2022) P02035)

- 2. Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab** Apr 2021
Muon g-2 Collaboration
arxiv.org/abs/2104.03240 (Phys.Rev.Accel.Beams 24 (2021) 044002)
- 1. Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm** Apr 2021
Muon g-2 Collaboration
arxiv.org/abs/2104.03281 (Phys.Rev.Lett. 126 (2021) 141801)

Ph.D. Thesis

- 1. Unifying Searches for New Physics with Precision Measurements of the W Boson Mass** 2024
D. Sathyan
drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9 (Ph.D. Thesis)